

Credit Card Approval Prediction

Phase 4: Project Planning Phase

SmartBridge / SkillWallet Virtual Internship Program

Team Member	Role
Tupakula Rishmitha Sri	Team Lead
Tejomai Kondapureddy	Member
Korrapolu Lahari	Member
Saragada Tirupathi Venkata Mohan Reddy	Member
Vijay Kanth Kalvakuntla	Member

1. Epics & User Stories

The project was broken into five epics, each containing user stories with story point estimates for sprint planning.

Epic 1: Data Collection

Story	Points
Download the application_record.csv and credit_record.csv datasets and prepare them for analysis.	2

Epic 2: Visualizing and Analysing the Data

Story	Points
Import required Python libraries for data analysis, visualization, and ML.	1
Read and explore the dataset to understand structure, features, and target variable.	2
Perform univariate analysis to examine the distribution of individual variables.	3
Conduct multivariate analysis to identify relationships among features.	3
Perform descriptive statistical analysis to summarize dataset characteristics.	2

Epic 3: Data Pre-Processing

Story	Points
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Identify and remove duplicate records.	2
Detect and handle missing values.	2
Clean and merge application_record.csv with aggregated credit_record.csv.	3
Apply feature engineering (EMI Paid Off, EMI Pastdues, Number of Loans).	5
Convert categorical variables into numerical form using LabelEncoder.	2

Epic 4: Model Building

Story	Points
Train a Logistic Regression model and evaluate performance.	3
Build and test a Decision Tree model.	3
Build and test a Random Forest model.	3
Build and test an XGBoost model.	3
Compare all models and select the best-performing one for deployment.	2

Epic 5: Application Building

Story	Points
Design and develop the HTML form for applicant input.	3
Build the Flask application and integrate the trained model.	5
Run, test, and validate the application end-to-end.	3

2. Sprint Schedule

Sprint	Duration	Focus	Story Points
Sprint 1	Week 1-2	Data Collection, EDA (Epics 1-2)	13
Sprint 2	Week 3-4	Data Preprocessing & Feature Engineering (Epic 3)	14
Sprint 3	Week 5-6	Model Building & Comparison (Epic 4)	14
Sprint 4	Week 7-8	Flask Application & Testing (Epic 5)	11

3. Team Roles & Responsibilities

Team Member	Primary Responsibility
Tupakula Rishmitha Sri (Team Lead)	Sprint planning, task allocation, and overall project coordination.
Tejomai Kondapureddy (Member)	Data collection and exploratory data analysis.
Korrapolu Lahari (Member)	Data preprocessing and feature engineering.
Saragada Tirupathi Venkata Mohan Reddy (Member)	Model building, training, and evaluation.
Vijay Kanth Kalvakuntla (Member)	Flask application development and integration.